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ELECTRONIC COMPONENT AND MANUFACTURING METHOD THEREOF

Abstract

A structure including a metal-insulator-metal (MIM) capacitor is provided. The MIM capacitor includes a second electrode, a dielectric layer, a first electrode. The second electrode has a main portion, at least one branch portion extending from the main portion, and a plurality of twig portions, extending from the branch portion. The dielectric layer covers a lower surface of the main portion, and extends for completely covering outer surfaces of the branch portion and twig portions of the second electrode. The first electrode conformally covers the dielectric layer, wherein the first electrode and second electrode are physically and electrically separated by the dielectric layer.

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Background/Summary

BACKGROUND

[0001] Modern day integrated chips include various active devices and/or passive devices. Metal-insulator-metal (MIM) capacitors are a common type of passive device that is often integrated into the integrated chips. For example, the capacitor may be used in various radio frequency (RF) circuits (e.g., an oscillator, phase-shift network, filter, converter, etc.), memory devices, and as a decoupling capacitor in high power microprocessor units (MPUs).

[0002] Therefore, it is desirable to provide a MIM capacitor or a manufacturing method thereof with a large capacitance and a small chip area requirement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. **1**A-**1**I illustrate various cross-sectional views of some embodiments of a method of forming an electronic component.

[0005] FIG. **1**J illustrates a cross-sectional view of some embodiments of an electronic component.

[0006] FIG. **1**K illustrates a top view of some embodiments of an electronic component.

[0007] FIG. **2** illustrates a various cross-sectional view of some embodiments of an electronic component.

[0008] FIG. **3** illustrates a various cross-sectional view of some embodiments of an electronic component.

[0009] FIG. **4** illustrates a various cross-sectional view of some embodiments of an electronic component.

[0010] FIG. **5** illustrates a various cross-sectional view of some embodiments of an electronic component.

[0011] FIG. **6** illustrates a flow diagram of some embodiments of a method of forming an electronic component.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the component in use or operation in addition to the

orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
[0014] It will be understood that, although the terms “first”, “second”, “third” and the like, may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element without departing from the scope of protection of the inventive concept.
[0015] FIGS. **1**A-**1**J illustrate various cross-sectional views of some embodiments of a method of forming an electronic component.
[0016] As shown in cross-sectional view of FIG. **1**A, a structure **100**A including a substrate **181** and a plurality of layers (e.g., layers **121**, **111**, **112**, **113**) or regions disposed on the substrate **181** or embedded in the substrate **181** is provided. The plurality of layers or regions may be formed by a suitable process such as low pressure chemical vapor deposition (LPCVD), plasma enhanced chemical vapor deposition (PECVD), high density chemical plasma vapor deposition (HDPCVD), atomic layer deposition (ALD), physical vapor deposition (PVD) or high density plasma chemical vapor deposition (HDPCVD), vapor transport deposition (VTD), ion implantation process, diffusion process, oxidation process, or the like. The plurality of layers or regions disposed on the substrate **181** or embedded in the substrate **181** may be considered a portion of the substrate **181**. Additionally, the substrate **181** as shown in the drawings may be illustrated as an example.
[0017] The substrate **181** may include a semiconductor substrate (e.g., a silicon (Si) substrate or a semiconductor wafer), a printed circuit board (e.g., an FR-4 printed circuit board), a glass substrate, a ceramic substrate, or a polymer substrate, but the disclosure is not limited thereto. A structure including a semiconductor substrate may be referred as a semiconductor structure. Take a semiconductor wafer as an example, the substrate **181** includes a crystalline silicon wafer. The substrate **181** may include various doped regions depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, the doped regions may be doped with p-type or n-type dopants. The doped regions may be doped with p-type dopants, such as boron or BF.sub.2; n-type dopants, such as phosphorus or arsenic; and/or combinations thereof. The doped regions may be configured for n-type Fin-type Field Effect Transistors (FinFETs) and/or p-type FinFETs. A plurality of layers or regions disposed on the substrate **181** (e.g., a semiconductor substrate) or embedded in the substrate may be considered a portion of the substrate. In some alternative embodiments, the substrate **181** is made of some other suitable elemental semiconductor, such as diamond or germanium; a suitable compound semiconductor, such as gallium arsenide, silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. The substrate **181** may further include interconnect structure formed over and electrically connected to the various doped regions. The interconnect structure may include a circuitry fabricated by front end of line (FEOL) or middle end of line (MEOL) processes. For example, an electrically conductive via may penetrate a dielectric region and electrically connect to a gate electrode, a source region, or a drain region.
[0018] Layers disposed on the substrate **181** include a first insulating layer **111**, a plurality of second insulating layer **112**, and a plurality of third insulating layer **113**. The second insulating layers **112** and the third insulating layers **113** are alternately stacked to form a corresponding stacked structure **171**A, and each third insulating layer **113** is sandwiched between two adjacent second insulating layers **112**. A material of the first insulating layer **111**, a material of the second insulating layer **112**, and a material of the third insulating layer **113** are different from each other. The first insulating layer **111** may be referred as an etching stop layer, and/or a material of the first insulating layer **111** may include silicon carbide (e.g., SiC). The second insulating layer **112** may be referred as a plasma enhanced oxide (PEOX) layer, and/or a material of the second insulating layer

112 may include silicon carbon oxide (SiCO/SiOC). The third insulating layer **113** may be referred as a sacrificial layer, and/or a material of the third insulating layer **113** may include silicon oxide (SiO) or silicon nitride (SiN).

[0019] As shown in the cross-sectional view of FIG. **1**B, a structure **100**B including a stacked structure **171**B is formed. The stacked structure **171**B as shown in FIG. **1**B may be formed by performing an etching process on the structure (e.g., the stacked structure **171**A or a structure including thereof) as shown in FIG. **1**A. Additionally, for simply or clearly, a certain layer (e.g., the second insulating layer **112** or the third insulating layer **113**, but not limited) may have a same reference sign in the different drawings before and after being patterned.

[0020] The stacked structure **171**B includes alternately stacked second insulating layers **112** and third insulating layers **113**. A three-dimensional shape of the stacked structure **171**B may be a cone or a pyramid. In a cross-section view (e.g., the drawing as shown in FIG. **1**B), an outline of the stacked structure **171**B is a trapezoid substantially. Moreover, in a top view, a corresponding surface area of each layer of the plurality of third insulating layers **113** may gradually increase toward the direction approaching the substrate **181**.

[0021] As shown in the cross-sectional view of FIG. **1**C, a structure **100**C including a fourth insulating layer **114** is formed. A material of the fourth insulating layer **114** is different from the material of the third insulating layer **113**. The fourth insulating layer **114** may be formed by an appropriate deposition process, then a planarization process (e.g., a chemical mechanical polishing (CMP) process) may be performed for forming a corresponding flat top surface.

[0022] The material of the fourth insulating layer **114** may be the same or similar to the material of the second insulating layer **112**. Therefore, there may be no clear interface between the second insulating layer **112** and the fourth insulating layer **114**. Structurally, the second insulating layer **112** and the fourth insulating layer **114** with a same or similar material may be regarded as a same insulating layer and may be referred as a fifth insulating layer **115**. That is, the plurality of third insulating layers **113** may be regarded as parallelly and respectively embedded in the fifth insulating layer **115**. A planarization process (e.g., a chemical mechanical polishing (CMP) process) may be performed to the fifth insulating layer **115** (e.g., the topmost second insulating layer **112** or a portion of fourth insulating layer **114** disposed thereon) for forming a corresponding flat top surface **115***a*.

[0023] As shown in the cross-sectional view of FIG. **1**D, a structure **100**D including at least one vertical trench **131** is formed. The vertical trench **131** may be formed by an appropriate etching process. The vertical trench **131** vertically penetrates through the fifth insulating layer **115** and the third insulating layers **113** embedded therein. A portion of the first conductive layer **121** disposed between the fifth insulating layer **115** and the substrate **181** is exposed by the vertical trench **131**. Moreover, a portion of each third insulating layers **113** is laterally exposed by the vertical trench **131**. The pattern of the vertical trench **131** is not limited in the disclosure. For example, in a top view, the pattern of the vertical trench **131** may be quadrilateral (e.g., a rectangle or a square) or circular.

[0024] As shown in the cross-sectional view of FIG. **1**E, a structure **100**E including a plurality of horizontal trenches **132** horizontally extending from the vertical trench **131** into the fifth insulating layer **115** is formed. The third insulating layers **113** (as shown in FIG. **1**D) laterally exposed by the vertical trench **131** are removed by a wet etching process to form the horizontal trenches **132** (as shown in FIG. **1**E). Thereby, the contour of the horizontal trenches **132** substantially corresponds to the shape of the third insulating layers **113**.

[0025] As shown in the cross-sectional view of FIG. **1**F, a structure **100**F including a second conductive layer **122** is formed. The second conductive layer **122** is formed by an appropriate deposition process, for example, including an ALD process. The second conductive layer **122** is disposed on the top surface **115***a* of the fifth insulating layer **115** and extended into the vertical trench **131**, and is further extended into the horizontal trenches **132** extending from the vertical trench **131** and disposed on the portion of the first conductive layer **121** exposed by the vertical trench **131**. The second conductive layer **122** may be conformally cover a portion of the outside surface of the fifth insulating layer **115** and the portion of the first conductive layer **121** exposed by the vertical trench **131**. That is, the second conductive layer **122** may not fill the entire vertical trench **131** and the entire horizontal trenches **132** essentially.

[0026] The second conductive layer **122** may be a stacked film structure of a plurality of conductive layers of different materials. For example, the second conductive layer **122** may be a stacked film structure of a metal nitride (e.g., tantalum nitride (TaN), titanium nitride (TiN), tungsten nitride (e.g., W_2N, WN, or WN_2), ruthenium nitride (RuN), or a stack or composition thereabove) layer, a metal (e.g, tantalum (Ta), titanium (Ti), ruthenium (Ru), tungsten (W), iridium (Ir), platinum (Pt), or a stack or composition thereabove) layer and/or a metal nitride layer. A metal nitride layer may be referred as a conductive glue layer or a barrier layer.

[0027] Depositing the second conductive layer **122** may include a conformal deposition process, for example, a CVD process, an ALD process, or the like, and/or a combination thereof. The conformal deposition process may include performing N deposition cycles until a desired thickness for the second conductive layer **122** is achieved, where N is any positive integer. In an embodiment, an optional inhibitor (e.g., a self-assembled monolayer (SAM) or small molecule inhibitor (SMI)) may be formed to cover the outer surfaces of the fifth insulating layer **115** and first conductive layer **121** exposed by the vertical trench **131** prior to conformal deposition process to further promote a substantially top-down directionality of a conformal deposition process. After the conformal deposition process is completed, an optional annealing process, for example at a temperature in a range of 200° C. to 600° C. in an inert gas (e.g., Ar, He, N_2, or the like) environment, may be performed. Then, a precursor may be provided over the fifth insulating layer **115**, and the precursor flows into the vertical trench **131** and further into the horizontal trenches **132**. The precursor may be selected from metal organic compounds, metal halides, metal carbonyls, metal complex, or the like with a relatively sticking coefficient. For example, the precursor may be a metal halide. In an embodiment where the target metal layer is a TiN layer, the precursor may be $TiCl_4$. Due to the relatively corresponding sticking coefficient of the precursor, the first precursor may first accumulate and attach to the top surface **115***a* of the fifth insulating layer **115**, then an upper sidewall of the vertical trench **131** and an upper horizontal trench **132**, and further then a lower sidewall of the vertical trench **131** and a lower horizontal trench **132**. That is, a surface concentration or density of the precursor may decrease along diffusion path thereof. For example, a surface concentration or density of the precursor may substantially decrease along a direction D**1** form the top of the vertical trench **131** towards the bottom of the vertical trench **131**. As such, a deposition thickness of the second conductive layer **122** may decrease along the direction D**1**. For example, a deposition thickness of the second conductive layer **122** on the sidewall of the upper horizontal trench **132** is thicker than a deposition thickness of the second conductive layer **122** on the sidewall of the lower horizontal trench **132**.

[0028] As shown in the cross-sectional view of FIG. **1**G, a structure **100**G including a dielectric layer **116** is formed. The dielectric layer **116** is formed by an appropriate deposition process, for example, an ALD process. The dielectric layer **116** is disposed on the outside surface of the second conductive layer **122**. The dielectric layer **116** may be conformally cover a portion of the outside surface of the second conductive layer **122**. That is, the dielectric layer **116** may not fill the entire vertical trench **131** and the entire horizontal trenches **132** (as labeled in FIGS. **1**E and **1**F, and for simply or clearly, skipping labeled in FIG. **1**G) essentially.

[0029] The dielectric layer **116** may be a stacked film structure of a plurality of dielectric layers of different materials. For example, the dielectric layer **116** may be a stacked film structure of high-K dielectric material layers. The high-K dielectric material may, for example, include silicon nitride (SiN; e.g., Si_3N_4), aluminum oxide (AlO; e.g., Al_2O_3), hafnium silicates (HfSiON), tantalum oxide (TaO; e.g., Ta_2O_5), zirconium oxide (ZrO; e.g., ZrO_2),

hafnium oxide (HfO; e.g., HfO.sub.2), titanium oxide (TiO; e.g., TiO.sub.2), barium strontium titanate oxide (BSTA; e.g., Ba.sub.0.8Sr.sub.0.2TiO.sub.3), strontium titanate oxide (STO; e.g., SrTiO.sub.3), or combinations thereof. In an embodiment, the dielectric layer **116** includes a laminate such as a zirconium oxide-aluminum oxide-zirconium oxide (e.g., ZrO.sub.2—Al.sub.2O.sub.3—ZrO.sub.2) laminate, sometimes referred to as a ZAZ laminate.
[0030] As shown in the cross-sectional view of FIG. **1**H, a structure **100**H including a third conductive layer **123** is formed. The third conductive layer **123** is formed by an appropriate process, for example, a deposition process (e.g., an ALD process) and/or a planting process (e.g., an electroplating process). The third conductive layer **123** and second conductive layer **122** are physically and electrically separated by the dielectric layer **116**.
[0031] The third conductive layer **123** is disposed on the top surface of the fifth insulating layer **115** and extended into the vertical trench **131** (as shown in FIG. **1**G), and is further extended into the horizontal trenches **132** extending from the vertical trench **131** (as shown in FIG. **1**G). Except for the space occupied by the second conductive layer **122** and the dielectric layer **116**, the third conductive layer **123** may substantially fill other remaining spaces of the vertical trench **131** and the horizontal trenches **132**, but the disclosure is not limited thereto.
[0032] The material of a portion of the third conductive layer **123** may be the same or similar to the material of the second conductive layer **122**. For example, the material of the seed layer of the third conductive layer **123** may be the same or similar to the material of the second conductive layer **122**.
[0033] Additionally, a planarization process (e.g., a chemical mechanical polishing (CMP) process) may be further performed on the third conductive layer **123** for forming a corresponding flat top surface, but the disclosure is not limited thereto.
[0034] As shown in the cross-sectional view of FIG. **1**I, a structure **100**I including a capacitor **140** is formed. For example, the structure **100**H as shown in FIG. **1**H is performed to an appropriate patterning process to form the structure **100**I including a capacitor **140** as shown in FIG. **1**I. The aforementioned patterning process may include a lithography and etching process to remove a portion of the second conductive layer **122**, a portion of the dielectric layer **116**, and a portion of the third conductive layer **123**, to respectively form a first electrode **141**, an insulator **142**, and a second electrode **143** of the capacitor **140**. That is, the first electrode **141** corresponds to a portion of the second conductive layer **122**, the insulator **142** corresponds to a portion of the dielectric layer **116**, and the second electrode **143** corresponds to a portion of the third conductive layer **123**. The capacitor **140** may be referred as a metal-insulator-metal (MIM) capacitor.
[0035] One or more insulating or dielectric layers (e.g., layers **145**, **146**, **147**, **148**) may be formed on the capacitor **140** for performing a subsequent processing. The aforementioned insulating or dielectric layers **145**, **146**, **147**, **148** may include a ZAZ laminate, a SiO layer, a SiN layer, a SiC layer, a SiOC layer, a SiON layer, a SiCN layer, a SiOCN layer, a undoped silicate glass (USG) layer, and/or a combination or stack thereof, but the disclosure is not limited thereto. At least one of the layers **145**, **146**, **147**, **148** may be referred as a high-K dielectric layer, at least one of the layers **145**, **146**, **147**, **148** may be referred as a barrier layer, at least one of the layers **145**, **146**, **147**, **148** may be referred as an etching stop layer, and/or at least one of the layers **145**, **146**, **147**, **148** may be referred as a capping layer, but the disclosure is not limited thereto.
[0036] FIG. **1**J illustrates a cross-sectional view of some embodiments of an electronic component. FIG. **1**K illustrates a top view of some embodiments of an electronic component. FIG. **1**J may be a cross-sectional view corresponding to the J-J′ crossline as shown in FIG. **1**I. It is worth noting that the conductor is essentially not on the cross-sectional view corresponding to the J-J′ crossline as shown in FIG. **1**I, but in order to understand the relative relationship of the structure, the relative position of the conductor is still shown in FIG. **1**K.
[0037] Referring to FIGS. **1**J and **1**K, the electronic component **100**J includes a least one capacitor **140**. The capacitor **140** has a main portion **140**A, at least one branch portion **140**B, and a plurality of twig portions **140**C. The branch portion **140**B extends from the main portion **140**A in a first

direction D**1**. The twig portions **140**C extend from corresponding branch portion **140**B in a second direction D**2**. The main portion **140**A, the branch portion **140**B, and the twig portions **140**C all include corresponding first electrodes **141**, second electrodes **143**, and insulators **142** disposed between the corresponding first electrode **141** and the corresponding second electrode **143**. The first direction D**1** is not parallel to the second direction D**2**. For example, the first direction D**1** is substantially perpendicular to the second direction D**2**.

[0038] In an embodiment, the electronic component **100**J further includes an insulating layer **115**. The main portion **140**A is disposed on the insulating layer **115**. The branch portion **140**B vertically penetrates through the insulating layer **115**. The twig portions **140**C horizontally or laterally extend from the corresponding branch portion **140**B and are respectively embedded in the insulating layer **115**.

[0039] In an embodiment, the insulating layer **115** is a homogeneous material. The aforementioned homogeneous material means one material of uniform composition throughout or a material, consisting of a combination of materials, that cannot be disjointed or separated into different materials by mechanical actions such as breaking, shearing, cutting, sawing, and grinding. In an embodiment, the insulating layer **115** with a homogeneous material has no interface within formed due to different materials.

[0040] In an embodiment, the peripheral contours of the plurality of twig portions **140**C gradually increase away from the main portion **140**A. For example, a first twig portion **140**C**1**, a second twig portion **140**C**2**, and a third twig portion **140**C**3** extend horizontally from a same branch portion **140**B. A vertically distance between the main portion **140**A and the first twig portion **140**C**1** is smaller than a vertically distance between the main portion **140**A and the second twig portion **140**C**2**, and a vertically distance between the main portion **140**A and the second twig portion **140**C**2** is smaller than a vertically distance between the main portion **140**A and the third twig portion **140**C**3**. In a top view (e.g., as shown in FIG. **1**K), a peripheral contour C**3** of the third twig portion **140**C**3** is larger than a peripheral contour C**2** of the second twig portion **140**C**2**, and the peripheral contour C**2** of the second twig portion **140**C**2** is larger than a peripheral contour C**1** of the first twig portion **140**C**1**.

[0041] In an embodiment, in the top view (e.g., as shown in FIG. **1**K), the plurality of twig portions **140**C extending horizontally from a same branch portion **140**B are overlapped. For example, the first twig portion **140**C**1**, the second twig portion **140**C**2**, and the third twig portion **140**C**3** are overlapped.

[0042] In an embodiment, in the top view (e.g., as shown in FIG. **1**K), a peripheral contour CA of the main portion **140**A is larger than the peripheral contour of each of the twig portions **140**C.

[0043] In an embodiment, the first electrode **141** is formed by an atomic layer deposition (ALD) process. In an embodiment, the thickness of the first electrodes **141** of the plurality of twig portions **140**C gradually decrease away from the main portion **140**A. For example, the thickness T**1** of the first electrode **141** of the first twig portion **140**C**1** is larger than the thickness (for simply and clearly, not directly labeled in the drawing) of the first electrode **141** of the second twig portion **140**C**2**, and the thickness of the first electrode **141** of the second twig portion **140**C**2** is larger than the thickness T**3** of the first electrode **141** of the third twig portion **140**C.

[0044] In an embodiment, the electronic component **100**J further includes a substrate **181**. The capacitor **140** is disposed on the substrate **181**. A conductive layer **121** is disposed on the substrate **181**. A portion of the conductive layer **121** disposed between the capacitor **140** and the substrate **181** is electrically connected to the first electrode **141** of the capacitor **140**.

[0045] In an embodiment, the electronic component **100**J further includes a conductor **144**. The conductor **144** is disposed on the conductor **144** and penetrates through the insulating or dielectric layers **145**, **146**, **147**, **148** disposed on the capacitor **140** to be electrically connected to the second electrode **143** of the capacitor **140**. In an embodiment, the conductor **144** is embedded in the second electrode **143** of the main portion **140**A.

[0046] FIG. **2** illustrates a various cross-sectional view of some embodiments of an electronic component.
[0047] The electronic component **200** as shown in FIG. **2** may be a portion of a die, but the disclosure is not limited thereto. For example, the substrate **181** includes a first region R**1** and a second region R**2**. Considering to a corresponding structure, the first region R**1** may be referred as a capacitance area, and the second region R**2** may be referred as a device area.
[0048] In an embodiment, the capacitor **140** is within a back-end-of-line (BEOL) structure of the die. That is, the capacitor **140** is referred as a portion of the interconnect structure (e.g., a BEOL interconnect structure) **201**.
[0049] The interconnect structure may include one or more corresponding insulating or dielectric layers (e.g., layers **211**, **212**, **213**, **214**, **215**, **216**) and one or more corresponding conductive layers (e.g., layers **221**, **222**). A layout design of the conductive layers may be adjusted according to actual needs, and is not limited by the disclosure. A portion of the conductive layers may be a corresponding conductive line and referred as a Mx layer, where x is zero or any positive integer. A portion of the conductive layers may be a corresponding conductive via and referred as a Vy layer, where y is zero or any positive integer. In an embodiment, the conductor **144** is a conductive via and/or a portion of the interconnect structure **201**.
[0050] A portion **121***a* of the conductive layer **121** corresponding to the first region R**1** is electrically connected to the first electrode **141** of the capacitor **140**. A portion **121***b* of the conductive layer **121** corresponding to the second region R**2** and/or away from the capacitor **140** may be a portion of an interconnector to be electrically connected a source, a drain, or a gate of a transistor.
[0051] FIG. **3** illustrates a various cross-sectional view of some embodiments of an electronic component.
[0052] The electronic component **300** as shown in FIG. **3** may be a portion of a die or a panel (e.g., a display panel), but the disclosure is not limited thereto. In an embodiment where the electronic component **300** is a die, the substrate **181** is a semiconductor substrate. In an embodiment where the electronic component **300** is a portion of a panel (e.g., a display panel), the substrate **181** is a printed circuit board or a glass substrate.
[0053] The electronic component **300** may further include a thin film transistor (TFT). The thin film transistor **340** may be a bottom gate thin film transistor, a top gate thin film transistor, or a dual gate thin film transistor. For example, thin film transistor **340** as shown in FIG. **3** may be referred as a bottom gate thin film transistor. A structure with one thin film transistor (e.g., the thin film transistor **340**) being electrically connected to one capacitor (e.g., the capacitor **140**) may be referred as a 1T1C structure. In an embodiment, a structure with one or more thin film transistors being electrically connected to one or more capacitors includes, for example, a 2T1C structure, a 3T1C/2C structure, a 4T1C/2C structure, or the like.
[0054] In an embodiment, a portion of the conductor **144** and a portion of the source/drain of the thin film transistor **340** are of the same film layer, but the disclosure is not limited thereto.
[0055] In an embodiment, a portion **121***b* of the conductive layer **121** away from the capacitor **140** is a portion of the gate of the thin film transistor **340**, but the disclosure is not limited thereto.
[0056] FIG. **4** illustrates a various cross-sectional view of some embodiments of an electronic component.
[0057] The electronic component **400** as shown in FIG. **4** may be a portion of a die or a panel, but the disclosure is not limited thereto. The electronic component **400** may further include at least one thin film transistor (e.g., the thin film transistor **441** and/or the thin film transistor **442**). In an embodiment, a portion of the conductor **144** and a portion of the gate of the thin film transistor (e.g., the thin film transistor **441** and/or the thin film transistor **442**) are a portion of the same film layer, but the disclosure is not limited thereto. An electrode of the capacitor **140**, the thin film transistor **441** and/or the thin film transistor **442** is electrically connected to a corresponding

contact terminal **429**.

[0058] In an embodiment, the thin film transistor **441** includes an n-type channel and be referred as a n-type transistor. The thin film transistor **442** includes an p-type channel and be referred as a p-type transistor. The gates of the thin film transistor **441** and the thin film transistor **442** are electrically connected to each other, the drain of the thin film transistor **441** and the source the thin film transistor **442** are electrically connected to each other.

[0059] In an embodiment, the thin film transistor **441** and the thin film transistor **442** are integrated as an inverter (e.g., a NOT gate), but the disclosure is not limited thereto. For example, the gates of an n-type transistor and a p-type transistor are electrically connected to an input, the source of the n-type transistor is electrically connected to a Vss source via a contact terminal, the drain of the n-type transistor and the source of the p-type transistor are electrically connected to an output via a contact terminal, and the drain of the p-type transistor is electrically connected to a Vdd source via a contact terminal.

[0060] The thin film transistor **441**, the thin film transistor **442**, and other one or more transistors not shown in FIG. **4** may be integrated as an AND gate, a NAND gate, an OR gate, a NOR gate, an XOR gate, or an XNOR gate.

[0061] It is worth noting that conductors not connected in the drawings may be electrically connected through other conductors not shown or in cross-sections not shown. For example, the portion **121***b* of the conductive layer **121** away from the capacitor **140** and being electrically connected to thin film transistors **441**, **442** is electrically connected to the portion **121***a* of the conductive layer **121** overlapped the capacitor **140**.

[0062] In an embodiment, the contact terminals **429** includes a die pad and/or a controlled collapse of chip connection (C4) bump, but the disclosure is not limited thereto.

[0063] FIG. **5** illustrates a various cross-sectional view of some embodiments of an electronic component.

[0064] The electronic component **500** as shown in FIG. **5** may be a portion of a stacked die or a package including a plurality of dies, but the disclosure is not limited thereto. For example, the electronic component **500** includes dies **501**, **502** which are stacked. The BEOL structure of the first die **501** and the BEOL structure of the second die **502** may be bonded face to face. At least one of the BEOL structures include a corresponding capacitor **140**.

[0065] The electronic component **500** may be heterogeneous integration with a plurality of dies. For example, the die **501** is a system-on-Chip (SoC) die, and the die **502** is an application specific integrated circuit (ASIC) die. In an embodiment, the die **501** further includes a CMOS image sensor (CIS) pixel structure. A device within the CIS pixel structure is electrically connected to a corresponding circuit within the BEOL structure by an appropriate conductor (e.g., a through silicon via (TSV)). A pixel (may be referred as a pixel unit PU) include at least one first sub-pixel (e.g., a red pixel) SP**1**, at least one second sub-pixel (e.g., a green pixel) SP**2**, and/or at least one third sub-pixel (e.g., a blue pixel) SP**3**. A plurality of pixel units PU may be arranged in an array. One or more pixels PU may correspond or be electrically connected to one or more corresponding capacitors **140**.

[0066] FIG. **6** illustrates a flow diagram of some embodiments of a method of forming an electronic component.

[0067] At act **601**, a stacked structure including alternately stacked second insulating layers and third insulating layers is provided. A material of the second insulating layer and a material of the third insulating layer are different from each other. FIGS. **1**A, **1**B, and/or **1**C illustrate cross-sectional views corresponding to various embodiments of act **601**. The stacked structure may be further disposed on a first conductive layer.

[0068] In an embodiment, the stacked structure is disposed on a substrate. An area of the topmost surface (e.g., the surface furthest away from the substrate) of the stacked structure is greater than an area of the bottommost surface (e.g., the surface closest to the substrate) of the stacked structure, and in a top view, the area of the topmost surface is overlapped and within the area of the bottommost surface. That is, in a cross-section view (e.g., the drawing as shown in FIG. **1**B), an outline of the stacked structure may be a trapezoid. FIG. **1**B illustrates a cross-sectional view corresponding to various embodiments of act **601**.

[0069] In an embodiment, the stacked structure is embedded in a fourth insulating layer. The second insulating layer and the fourth insulating layer with a same or similar material may be regarded as a same insulating layer and may be referred as a fifth insulating layer. FIG. **1**C illustrates a cross-sectional view corresponding to various embodiments of act **601**.

[0070] At act **602**, a removal process is performed for removing a portion of the fifth insulating layer (or, a portion of the second insulating layer) and the third insulating layer to form at least one first trench (e.g., the vertical trench **131**) and a plurality of second trenches (e.g., the horizontal trenches **132**) laterally or horizontally extending from the first trench. FIG. **1**E illustrates a cross-sectional view corresponding to various embodiments of act **602**. The first trench may be further exposed a portion of the first conductive layer.

[0071] In an embodiment, a first removal process is performed for removing a portion of the fifth insulating layer (or, a portion of the second insulating layer), and a portion of the third insulating layer to form at least one first trench. Then, a second removal process is performed for removing the remained third insulating layer to form a plurality of second trenches horizontally extending from the first trench. FIGS. **1**D-**1**E illustrate cross-sectional views corresponding to various embodiments of act **602**.

[0072] At act **603**, after performing the removal process, an appropriate deposition process is performed for forming a second conductive layer disposed on an outer surface of the fifth insulating (or, a portion of the second insulating layer). FIG. **1**F illustrates a cross-sectional view corresponding to various embodiments of act **603**. The second conductive layer may be further disposed on the portion of the first conductive layer exposed by the first trench.

[0073] In an embodiment, the second conductive layer is formed including an ALD process.

[0074] In an embodiment, the second conductive layer is conformally cover a portion of the outside surface of the fifth insulating layer and the portion of the first conductive layer exposed by the first trench. In an embodiment, the second conductive layer is not fill the entire first trench and the entire second trenches essentially.

[0075] At act **604**, after forming the second conductive layer, a dielectric layer disposed on the second conductive layer is formed. FIG. **1**G illustrates a cross-sectional view corresponding to various embodiments of act **604**.

[0076] In an embodiment, the dielectric layer is conformally cover a portion of the outside surface of the second conductive layer. In an embodiment, the dielectric layer may not fill the entire first trench and the entire second trenches essentially.

[0077] At act **605**, after forming the dielectric layer, a third conductive layer disposed on the dielectric layer is formed. The third conductive layer and second conductive layer are physically and electrically separated by the dielectric layer. FIG. **1**H or **1**I illustrate cross-sectional views corresponding to various embodiments of act **605**.

[0078] Accordingly, in some embodiments, the present disclosure relates to an electronic component including at least one capacitor. The electronic component may have a better heat dissipation. The capacitor includes a main portion, at least one branch portion, and a plurality of twig portion. The branch portion extends from the main portion in a first direction. The twig portions extend from corresponding branch portion in a first direction. The main portion, the branch portion, and the twig portions all include corresponding first electrodes, second electrodes, and insulators disposed between the first electrode and the second electrode. As such, the capacitor of the disclosure may have better or more capacitance. For example, in a top view, under the same unit area, the capacitor of the disclosure may have a higher capacitance value.

[0079] In accordance with some embodiments of the present disclosure, a structure comprises an insulating layer and a metal-insulator-metal (MIM) capacitor. The insulating layer is disposed on a substrate. The MIM capacitor is disposed on the substrate. The MIM capacitor comprises a main portion, at least one branch portion, and a plurality of twig portions. The main portion is disposed on the insulating layer. The branch portion extends from the main portion. The twig portions extend from the branch portion and are respectively embedded in the insulating layer. In an embodiment, in a top view, peripheral contours of the twig portions gradually increase away from the main portion. In an embodiment, in a top view, a peripheral contour of the main portion is larger than peripheral contours of the twig portions. In an embodiment, adjacent two of the twig portions are electrically connected only through the branch portion. In an embodiment, a portion of the insulating layer is disposed between adjacent two of the twig portions. In an embodiment, the insulating layer is a homogeneous material. In an embodiment, the branch portion penetrates through the insulating layer, and the twig portions laterally extend from the branch portion. In an embodiment, the structure further comprises a conductor embedded in the main portion of the MIM capacitor. In an embodiment, the structure further comprises at least one thin film transistor (TFT) electrically connected to the MIM capacitor. In an embodiment, the structure further comprises at least one image sensor pixel electrically connected to the MIM capacitor.

[0080] In accordance with some embodiments of the present disclosure, a structure comprises a metal-insulator-metal (MIM) capacitor. The MIM capacitor comprises a second electrode, a dielectric layer, a first electrode. The second electrode has a main portion, at least one branch portion extending from the main portion, and a plurality of twig portions, extending from the branch portion. The dielectric layer covers a lower surface of the main portion, and extends for completely covering outer surfaces of the branch portion and twig portions of the second electrode. The first electrode conformally covers the dielectric layer, wherein the first electrode and second electrode are physically and electrically separated by the dielectric layer. In an embodiment, the first electrode is a structurally continuous film layer. In an embodiment, the twig portions of the second electrode are structurally respectively annularly extended from the branch portion. In an embodiment, the twig portions are different in sizes. In an embodiment, the structure further comprises a conductor embedded in the main portion of the second electrode.

[0081] In accordance with some embodiments of the present disclosure, a method comprises: providing a stacked structure including alternately stacked second insulating layers and third insulating layers, wherein a material of the second insulating layer and a material of the third insulating layer are different from each other; performing a removal process for removing a portion of the second insulating layer and the third insulating layer to form at least one first trench and a plurality of second trenches laterally extending from the first trench; forming a second conductive layer disposed on an outer surface of the second insulating layer; forming a dielectric layer disposed on the second conductive layer; and forming a third conductive layer disposed on the dielectric layer, wherein the third conductive layer and second conductive layer are physically and electrically separated by the dielectric layer. In an embodiment, in a top view, an area of a topmost surface of the stacked structure is overlapped and within an area of a bottommost surface of the stacked structure; the stacked structure is disposed on a substrate, the topmost surface is the surface of the stacked structure furthest away from the substrate, and the bottommost surface is the surface of the stacked structure closest to the substrate. In an embodiment, the method further comprises: forming a fourth insulating layer laterally covering the stacked structure, wherein a material of the fourth insulating layer is essentially the same as the material of the second insulating layer. In an embodiment, the method further comprises: forming a first conductive layer disposed on a substrate; and forming a first insulating layer disposed on the first conductive layer; wherein the material of the second insulating layer, the material of the third insulating layer, and a material of the first insulating layer are different from each other; a portion of the first insulating layer is removed by the removal process; and the second conductive layer is further disposed on a portion of the first conductive layer exposed by the first trench. In an embodiment, the removal process

comprises a first removal process and a second removal process, wherein the portion of the first insulating layer, the portion of the second insulating layer, and a portion of the third insulating layer are removed by the first removal process for forming the first trench; and a remained portion of the third insulating layer are removed by the second removal process for forming the second trenches.
[0082] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A structure, comprising: an insulating layer disposed on a substrate; and a metal-insulator-metal (MIM) capacitor disposed on the substrate, wherein the MIM capacitor comprises: a main portion, disposed on the insulating layer; at least one branch portion, extending from the main portion; and a plurality of twig portions, extending from the branch portion and being respectively embedded in the insulating layer.
2. The structure of claim 1, wherein in a top view, peripheral contours of the twig portions gradually increase away from the main portion.
3. The structure of claim 1, wherein in a top view, a peripheral contour of the main portion is larger than peripheral contours of the twig portions.
4. The structure of claim 1, wherein adjacent two of the twig portions are electrically connected only through the branch portion.
5. The structure of claim 1, wherein a portion of the insulating layer is disposed between adjacent two of the twig portions.
6. The structure of claim 1, wherein the insulating layer is a homogeneous material.
7. The structure of claim 1, wherein: the branch portion penetrates through the insulating layer; and the twig portions laterally extend from the branch portion.
8. The structure of claim 1, further comprising: a conductor, embedded in the main portion of the MIM capacitor.
9. The structure of claim 1, further comprising: at least one thin film transistor (TFT), electrically connected to the MIM capacitor.
10. The structure of claim 1, further comprising: at least one image sensor pixel electrically connected to the MIM capacitor.
11. A structure, comprising: a metal-insulator-metal (MIM) capacitor, comprising: a second electrode, having a main portion, at least one branch portion extending from the main portion, and a plurality of twig portions extending from the branch portion; a dielectric layer, covering a lower surface of the main portion, and extending for completely covering outer surfaces of the branch portion and twig portions of the second electrode; and a first electrode, conformally covering the dielectric layer, wherein the first electrode and second electrode are physically and electrically separated by the dielectric layer.
12. The structure of claim 11, wherein the first electrode is a structurally continuous film layer.
13. The structure of claim 11, wherein the twig portions of the second electrode are structurally respectively annularly extended from the branch portion.
14. The structure of claim 11, wherein the twig portions are different in sizes.
15. The structure of claim 11, further comprising: a conductor, embedded in the main portion of the second electrode.

16. A method, comprising: providing a stacked structure including alternately stacked second insulating layers and third insulating layers, wherein a material of the second insulating layer and a material of the third insulating layer are different from each other; performing a removal process for removing a portion of the second insulating layer and the third insulating layer to form at least one first trench and a plurality of second trenches laterally extending from the first trench; forming a second conductive layer disposed on an outer surface of the second insulating layer; forming a dielectric layer disposed on the second conductive layer; and forming a third conductive layer disposed on the dielectric layer, wherein the third conductive layer and second conductive layer are physically and electrically separated by the dielectric layer.
17. The method of claim 16, wherein in a top view, an area of a topmost surface of the stacked structure is overlapped and within an area of a bottommost surface of the stacked structure; wherein the stacked structure is disposed on a substrate, the topmost surface is the surface of the stacked structure furthest away from the substrate, and the bottommost surface is the surface of the stacked structure closest to the substrate.
18. The method of claim 17, further comprising: forming a fourth insulating layer laterally covering the stacked structure, wherein: a material of the fourth insulating layer is essentially the same as the material of the second insulating layer.
19. The method of claim 16, further comprising: forming a first conductive layer disposed on a substrate; and forming a first insulating layer disposed on the first conductive layer; wherein: the material of the second insulating layer, the material of the third insulating layer, and a material of the first insulating layer are different from each other a portion of the first insulating layer is removed by the removal process; and the second conductive layer is further disposed on a portion of the first conductive layer exposed by the first trench.
20. The method of claim 19, wherein the removal process comprises a first removal process and a second removal process, wherein: the portion of the first insulating layer, the portion of the second insulating layer, and a portion of the third insulating layer are removed by the first removal process for forming the first trench; and a remained portion of the third insulating layer are removed by the second removal process for forming the second trenches.